

COMMISSIONLENS - Commission-Adjusted Alpha Prediction in Mutual Funds

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1. Introduction

Indian mutual fund investors often choose regular plans without realizing the long-term impact of distributor commissions embedded in higher expense ratios. Although direct and regular plans invest in the same underlying portfolio, regular plans charge an additional 0.5%–1.5% annually.

Over long investment horizons, this cost compounds significantly and may reduce investor wealth substantially. The key financial question is:

Are these commissions justified by superior future performance?

CommissionLens addresses this problem using machine learning and benchmark-adjusted alpha prediction.

2. Objective

The project aims to predict whether a mutual fund will generate sufficient future alpha to justify its commission cost.

The core metric used is:

- $\text{Net Alpha} = \text{Fund Return} - \text{Benchmark Return} - \text{Expense Gap}$

A fund is considered:

- **Commission Justified** if future net alpha > 0
 - **Commission Unjustified** otherwise
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3. Dataset Construction

A large-scale Indian mutual fund dataset was built using:

- AMFI API
- mfapi.in
- NIFTY 50 benchmark data

Dataset Summary

Component	Value
Mutual fund schemes	766
NAV observations	1.48M+
Quarterly ML observations	17,868

4. Feature Engineering

The following rolling financial indicators were engineered:

Feature	Purpose
Rolling Return	Historical performance
Volatility	Risk estimation
Sharpe Ratio	Risk-adjusted return
Beta	Market sensitivity
Alpha	Benchmark-adjusted excess return
Information Ratio	Consistency of outperformance
Momentum	Trend persistence

5. Machine Learning Pipeline

```
=====
XGBoost
=====
Accuracy: 0.6925
Precision: 0.7028
Recall: 0.8908
F1 Score: 0.7857
ROC-AUC: 0.7151
Precision@TopDecile: 0.8375
```

```
=====
Random Forest
=====
Accuracy: 0.6402
Precision: 0.6382
Recall: 0.9960
F1 Score: 0.7780
ROC-AUC: 0.6824
Precision@TopDecile: 0.8291
```

```
=====
Logistic Regression
=====
Accuracy: 0.6273
Precision: 0.6337
Recall: 0.9744
F1 Score: 0.7679
ROC-AUC: 0.5989
Precision@TopDecile: 0.8235
```

Best Model — XGBoost

Metric	Score
Accuracy	69.25%
ROC-AUC	0.71
F1 Score	0.78
Precision@TopDecile	0.83

The most important metric was Precision@TopDecile, which measures:
Among the top 10% highest-confidence predictions, how many funds were actually commission-justified.
The model achieved approximately 83% precision in this category.

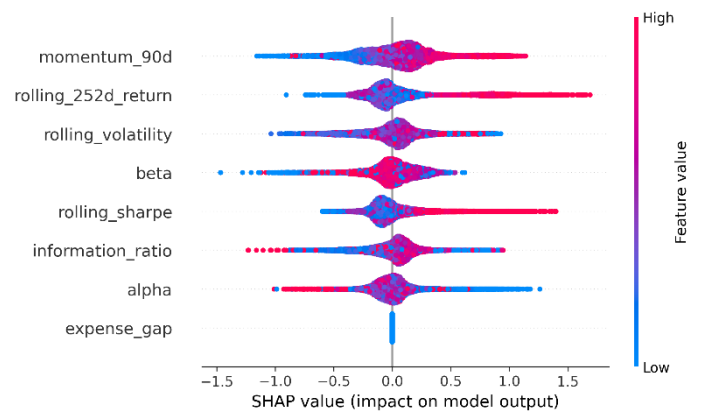
6. SHAP Explainability

SHAP analysis was used to interpret model behavior.

Most Important Predictive Features

Rank Feature

- 1 Momentum (90D)
- 2 Rolling 252D Return
- 3 Rolling Volatility
- 4 Beta
- 5 Sharpe Ratio



Key Insight

Funds with:

- stronger momentum,
- stable risk-adjusted returns,
- lower volatility,
- persistent benchmark outperformance

were more likely to justify higher commissions.

